

CSCI 461 – Introduction to Big Data

Assignment #2 – Report

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Introduction

Assignment 2 consists of two major components: MongoDB queries and MapReduce implementation. They focus on different skills and . This report is a thorough guide for the steps we followed and the results we got.

Part 1: MongoDB

This part covers all the work done using MongoDB. The data from mds.json was imported into a MongoDB container and a set of queries were written to answer the required questions.

1. Setting Up the Database

Starting the MongoDB Container

The MongoDB container was started using Docker. The mds.json file was then copied into the container and imported into the moviesDB database under a collection called moviesColl. The command used is shown in the screenshot below. A total of 1401 documents were imported successfully.

```
C:\Users\DELL>docker start mongo
mongo

C:\Users\DELL>cd /d "D:\^^-university-^^\current\big-data\assignment-2"

D:\^^-university-^^\current\big-data\assignment-2>docker cp mds.json mongo:/tmp/
Successfully copied 2.3MB (transferred 2.3MB) to mongo:/tmp/

D:\^^-university-^^\current\big-data\assignment-2>docker exec -it mongo mongoimport --jsonArray --db moviesDB --collection moviesColl --file /tmp/mds.json
2026-05-11T11:13:02.195+0000    connected to: mongoddb://localhost/
2026-05-11T11:13:02.285+0000    1401 document(s) imported successfully. 0 document(s) failed to import.

What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug mongo
  Learn more at https://docs.docker.com/go/debug-cli/

D:\^^-university-^^\current\big-data\assignment-2>
```

Connecting to MongoDB with mongosh

After the container was running, mongosh was used to connect to MongoDB. The screenshot below shows the connection being established successfully.

```
D:\^^-university-^^\current\big-data\assignment-2>docker exec -it mongo mongosh
Current Mongosh Log ID: 6a01b9eb51b99f915544ba88
Connecting to:      mongoddb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.2
Using MongoDB:      8.2.7
Using Mongosh:      2.8.2

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-05-11T11:09:44.705+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
  2026-05-11T11:09:45.107+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
  2026-05-11T11:09:45.107+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
  2026-05-11T11:09:45.107+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
  2026-05-11T11:09:45.107+00:00: We suggest setting the contents of sysfsFile to 0.
  2026-05-11T11:09:45.107+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
  -----

test>
```

The moviesDB Database and moviesColl Collection

The screenshot below shows that the moviesDB database was created and that the moviesColl collection exists inside it.

```
PowerShell x mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 x + v
test> show dbs
admin      48.00 KiB
config     72.00 KiB
local      68.00 KiB
moviesDB   1.00 MiB
test> use moviesDB
switched to db moviesDB
moviesDB> show collections
moviesColl
moviesDB> |
```

2. MongoDB Queries

Query A – Indexing

An index was created on the name field so that searching by movie or show name would be faster. Then, a query was run to search for a specific title by name.

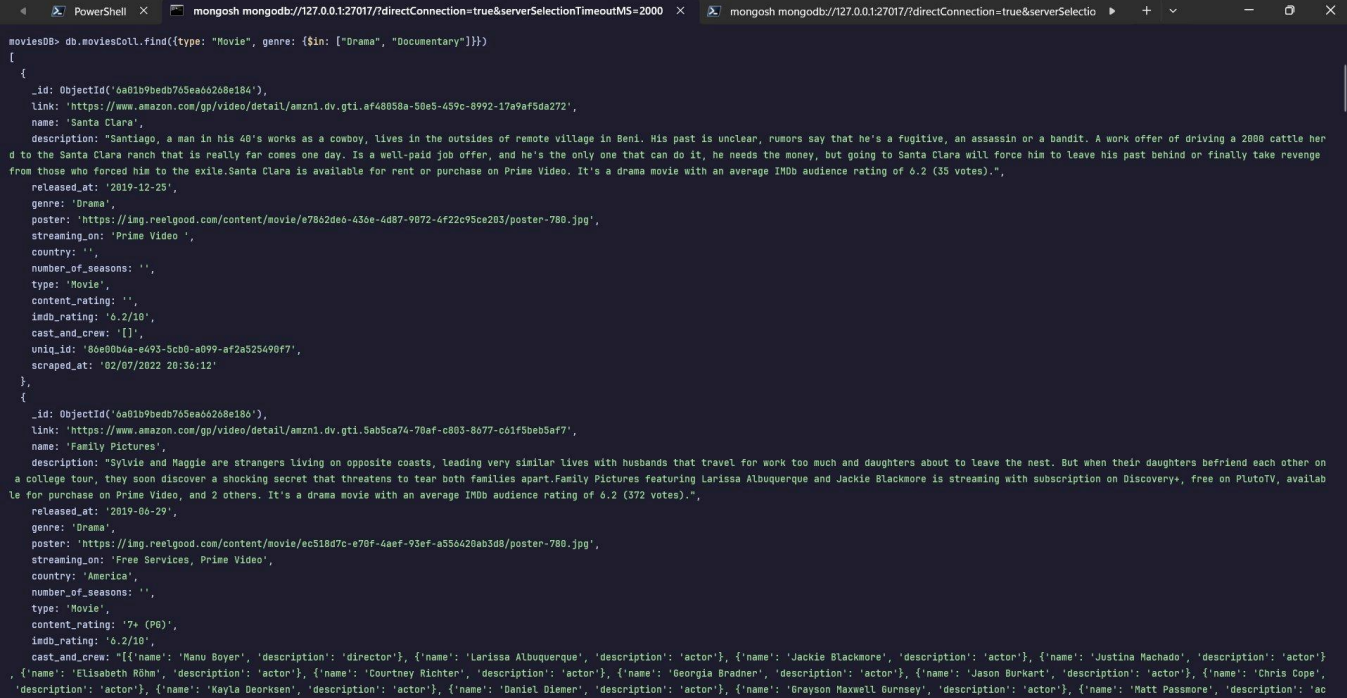
```
// Create an index on the name field, then search for a title by name
db.moviesColl.createIndex({ name: 1 })
db.moviesColl.find({ name: "Ice Airport Alaska" })
```

```
PowerShell x mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 x mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 x + v
moviesDB> db.moviesColl.createIndex({name: 1})
name_1
moviesDB> db.moviesColl.find({name: "Ice Airport Alaska"})
[
  {
    _id: ObjectId('6a01b9bedb765ea66268e183'),
    link: 'https://www.amazon.com/gp/video/detail/amzn1.dv.gti.fcbaefca-e87f-9adb-a197-9f951e3117bb',
    name: 'Ice Airport Alaska',
    description: "A chronicle of the daily challenges of the airport staff who manage to keep the Ted Stevens Anchorage International Airport (ANC) open, freight flowing, and passengers flying, all despite weather that regularly dips to -38 Fahrenheit and wind speeds that typically exceed 100 mph. Ice Airport Alaska has one or more episodes streaming on Paramount+, streaming with subscription on fuboTV, streaming with subscription on Smithsonian Earth (Via Prime Video), and 3 others. It's a documentary show with 12 episodes over 2 seasons. Ice Airport Alaska is still airing with no announced date for the next episode or season. It has a high IMDb audience rating of 7.8 (48 votes).",
    released_at: '2020-10-26',
    genre: 'Documentary',
    poster: 'https://img.reelgood.com/content/show/0045f9a9-943b-4dda-9a01-384d5bc51f3d/poster-780.jpg',
    streaming_on: 'Paramount+, Prime Video',
    country: '',
    number_of_seasons: '2',
    type: 'TV Show',
    content_rating: '',
    imdb_rating: '7.8/10',
    cast_and_crew: '[]',
    uniq_id: '39e5ebf0-3ccd-52e0-a9ed-79dcd7ba0591',
    scraped_at: '02/07/2022 20:36:09'
```

Query B – Logical Operations

This query finds all titles where the type is Movie and the genre is either Drama or Documentary.

```
// Find all Movies where the genre is Drama or Documentary
db.moviesColl.find({ type: "Movie", genre: { $in: ["Drama", "Documentary"] } })
```



```
moviesDB> db.moviesColl.find({type: "Movie", genre: {'$in': ["Drama", "Documentary"]}})
{
  {
    _id: ObjectId('6a01b9bedb705ea66268e184'),
    link: 'https://www.amazon.com/gp/video/detail/amzn1.dv.gti.af48058a-50e5-459c-8992-17a9af5da272',
    name: 'Santa Clara',
    description: 'Santiago, a man in his 40's works as a cowboy, lives in the outsid... of remote village in Beni. His past is unclear, rumors say that he's a fugitive, an assassin or a bandit. A work offer of driving a 2000 cattle herd to the Santa Clara ranch that is really far comes one day. Is a well-paid job offer, and he's the only one that can do it, he needs the money, but going to Santa Clara will force him to leave his past behind or finally take revenge from those who forced him to the exile.Santa Clara is available for rent or purchase on Prime Video. It's a drama movie with an average IMDb audience rating of 6.2 (35 votes).',
    released_at: '2019-12-25',
    genre: 'Drama',
    poster: 'https://img.reelgood.com/content/movie/e7862de0-436e-4d87-9072-4f22c95ce203/poster-780.jpg',
    streaming_on: 'Prime Video ',
    country: '',
    number_of_seasons: '',
    type: 'Movie',
    content_rating: '',
    imdb_rating: '6.2/10',
    cast_and_crew: [],
    uniq_id: '86e8084e-e493-5cb0-a099-af2a525490f7',
    scrapped_at: '02/07/2022 20:36:12'
  },
  {
    _id: ObjectId('6a01b9bedb705ea66268e186'),
    link: 'https://www.amazon.com/gp/video/detail/amzn1.dv.gti.5ab5ca74-70af-c803-8677-c61f5beb5af7',
    name: 'Family Pictures',
    description: 'Sylvie and Maggie are strangers living on opposite coasts, leading very similar lives with husbands that travel for work too much and daughters about to leave the nest. But when their daughters befriend each other on a college tour, they soon discover a shocking secret that threatens to tear both families apart.Family Pictures featuring Larissa Albuquerque and Jackie Blackmore is streaming with subscription on Discovery+, free on PlutoTV, available for purchase on Prime Video, and 2 others. It's a drama movie with an average IMDb audience rating of 6.2 (372 votes).',
    released_at: '2019-06-29',
    genre: 'Drama',
    poster: 'https://img.reelgood.com/content/movie/ec518d7c-e70f-4aef-93ef-a556420ab3d8/poster-780.jpg',
    streaming_on: 'Free Services, Prime Video',
    country: 'America',
    number_of_seasons: '',
    type: 'Movie',
    content_rating: '7+ (PG)',
    imdb_rating: '6.2/10',
    cast_and_crew: [{name: 'Manu Boyer', 'description': 'director'}, {name: 'Larissa Albuquerque', 'description': 'actor'}, {name: 'Jackie Blackmore', 'description': 'actor'}, {name: 'Justine Machado', 'description': 'actor'}, {name: 'Elisabeth R  hm', 'description': 'actor'}, {name: 'Courtney Richter', 'description': 'actor'}, {name: 'Georgia Bradner', 'description': 'actor'}, {name: 'Jason Burkart', 'description': 'actor'}, {name: 'Chris Cope', 'description': 'actor'}, {name: 'Kayla Deorksen', 'description': 'actor'}, {name: 'Daniel Diemer', 'description': 'actor'}, {name: 'Grayson Maxwell Gurnsey', 'description': 'actor'}, {name: 'Matt Passmore', 'description': 'ac
```

Query C – Aggregation

This aggregation pipeline counts how many titles exist for each type (Movie or TV Show) and sorts the result from highest to lowest count. The result shows 1232 Movies and 169 TV Shows.

```
// Count titles per type and sort from highest to lowest
db.moviesCol.aggregate([
  { $group: { _id: "$type", count: { $sum: 1 } } },
  { $sort: { count: -1 } }
])
```

```
moviesDB> db.moviesColl.aggregate([
|   {$group: {_id: "$type", count: {$sum: 1}}},
|   {$sort: {count: -1}}
| ])
[ { _id: 'Movie', count: 1232 }, { _id: 'TV Show', count: 169 } ]
moviesDB> 
```

Query D – Top 5 Highest Rated Movies

This query finds the top 5 highest-rated Movies released after 2010, available on Prime Video or Netflix, and that have an IMDb rating. The results are sorted by the numeric IMDb rating and only show the relevant fields. The query uses an aggregation pipeline to properly extract and compare the numeric rating value.

```
moviesDB> db.moviesColl.aggregate([
|   {$match: {
|     type: "Movie",
|     imdb_rating: {$exists: true, $nin: [""]},
|     released_at: {$exists: true, $nin: [""]},
|     streaming_on: {$regex: "Prime Video|Netflix", $options: "i"}
|   }},
|   {$project: {
|     name: 1,
|     genre: 1,
|     country: 1,
|     streaming_on: 1,
|     released_at: {
|       $toInt: {
|         $arrayElemAt: [
|           {$split: ["$released_at", "-"]},
|           0
|         ]
|       }
|     },
|     imdb_rating: {
|       $toDouble: {
|         $arrayElemAt: [
|           {$split: ["$imdb_rating", "/" ]},
|           0
|         ]
|       }
|     }
|   }},
|   {$match: {released_at: {$gt: 2010}}},
|   {$sort: {imdb_rating: -1}},
|   {$limit: 5}
| ])
|
| {
|   _id: ObjectId('6a81b9bedb765ea66268a5ec'),
|   name: 'Liquor House Comedy presents Brown Sugar Night',
|   genre: 'Documentary',
|   streaming_on: 'Free Services, Prime Video',
|   country: '',
| }
```

```
[
  {
    _id: ObjectId('6a01b9bed765ea66268e5ec'),
    name: 'Liquor House Comedy presents Brown Sugar Night',
    genre: 'Documentary',
    streaming_on: 'Free Services, Prime Video',
    country: '',
    released_at: 2020,
    imdb_rating: 9.7
  },
  {
    _id: ObjectId('6a01b9bed765ea66268e4ce'),
    name: 'Messiah',
    genre: 'Drama, Horror',
    streaming_on: 'Prime Video ',
    country: '',
    released_at: 2016,
    imdb_rating: 9.3
  },
  {
    _id: ObjectId('6a01b9bed765ea66268e308'),
    name: 'Why She Smiles',
    genre: 'Biography, Documentary',
    streaming_on: 'Free Services, Prime Video',
    country: '',
    released_at: 2021,
    imdb_rating: 9.1
  },
  {
    _id: ObjectId('6a01b9bed765ea66268e48f'),
    name: 'Attache',
    genre: 'Documentary',
    streaming_on: 'Prime Video ',
    country: '',
    released_at: 2018,
    imdb_rating: 9
  },
  {
    _id: ObjectId('6a01b9bed765ea66268e44f'),
    name: 'Kangozi Ki Kashi',
    genre: 'Documentary',
    streaming_on: 'Prime Video ',
    country: '',
    released_at: 2016,
    imdb_rating: 8.9
  }
]
moviesDB> |
```

3. Bonus – Advanced MongoDB Query

For the bonus, a \$bucket aggregation was used to group all titles by their IMDb rating ranges. This is an advanced MongoDB feature that was not used in any of the queries above. It groups documents into defined rating brackets and counts the titles falling into each one.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 × mongosh mongodb://127.0.0.1:27017/?directConnection=true&
moviesDB> db.moviesColl.aggregate([{$match: {imdb_rating: {$exists: true, $nin: [""]}}, {
|   $bucket: {
|     groupBy: {
|       $toDouble: {
|         $arrayElemAt: [
|           {$split: ["$imdb_rating", "/"]},
|           0
|         ]
|       }
|     },
|     boundaries: [0, 5, 7, 8, 10],
|     default: "Other",
|     output: {
|       count: { $sum: 1 }
|     }
|   }
| })
|
| { _id: 0, count: 226 },
| { _id: 5, count: 590 },
| { _id: 7, count: 288 },
| { _id: 8, count: 102 }
| ]
moviesDB> |
```

Part Two: Hadoop MapReduce

This part covers the MapReduce job written in Java. The mds.json file was first converted to CSV format, then copied to HDFS, and a MapReduce program was written in IntelliJ IDEA to calculate rating statistics per content type, genre, and release decade.

1. Converting JSON to CSV

The mds.json file was converted to a CSV file. The CSV file includes a header row with all field names. Array values such as genre were written as comma-separated values inside one cell. Null and empty values were kept as-is during conversion since data cleaning was handled inside the MapReduce job instead.

2. Copying the CSV File to HDFS

After the CSV file was ready, the Hadoop container was used to copy it to HDFS so the MapReduce job could read it as input.

3. The MapReduce Job – ARDriver.java

The MapReduce job was written in a single Java class called ARDriver.java. The class contains the Driver, Mapper, Reducer, and Combiner code all in one file.

What the Mapper Does

The Mapper reads each line from the CSV file and extracts the content type, the first genre (if more than one exists), and the release decade from the released_at field. It also extracts the numeric value from the imdb_rating field. Any record with a null, empty, or invalid value in genre, type, released_at, or imdb_rating is skipped entirely.

What the Reducer Does

The Reducer receives all values grouped by the key, which is a combination of content type, first genre, and release decade. For each group it calculates the total count of valid titles, the average IMDb rating, the minimum rating, and the maximum rating.

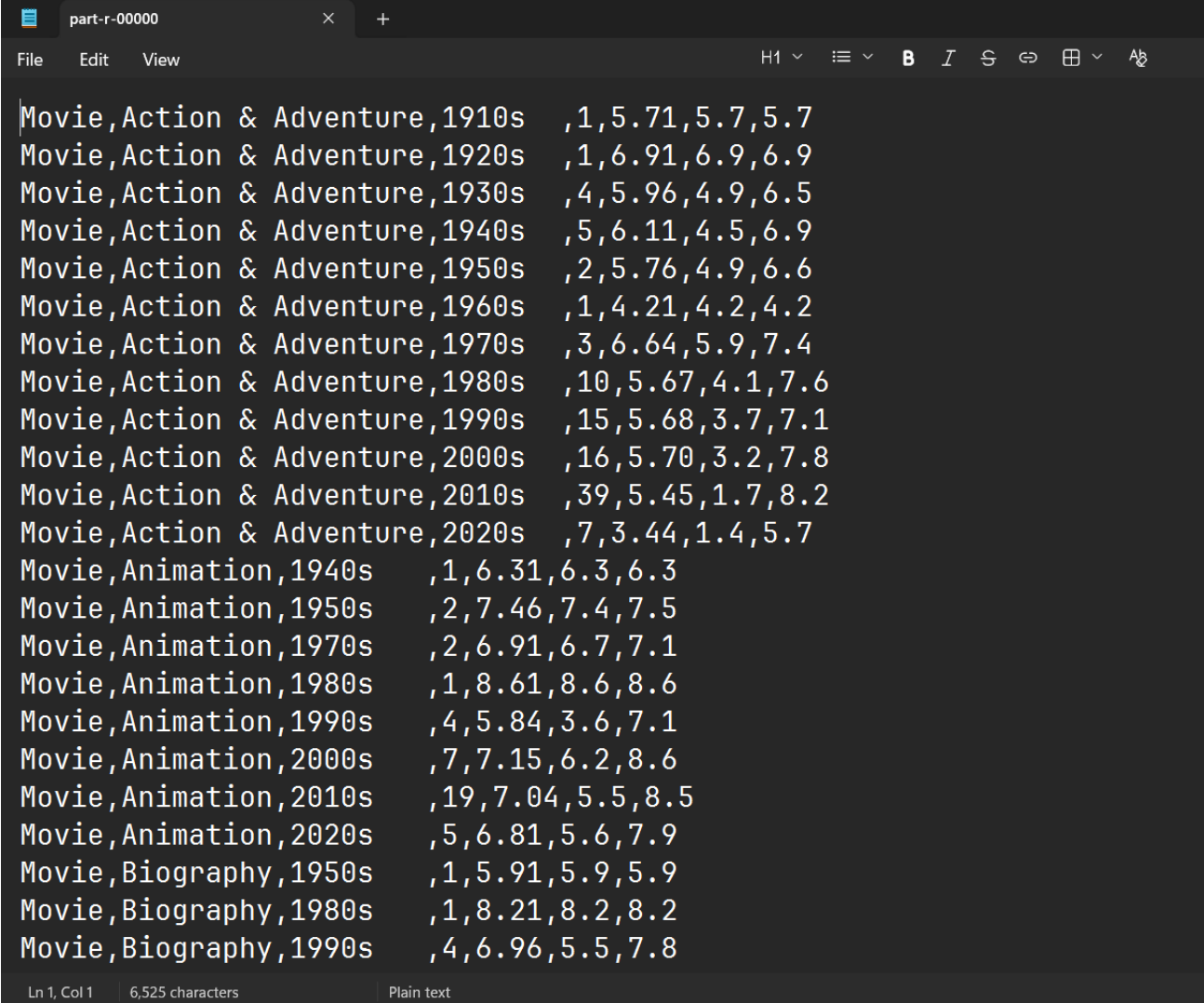
Bonus – Combiner

The Combiner was included inside ARDriver.java. It runs locally after the Mapper on each node and does a partial aggregation before the data is shuffled and sent to the Reducer.

This reduces the amount of intermediate data transferred over the network, which speeds up the job.

4. Output Format

The final output is saved in the file `part-r-00000`. Each line in the output shows the content type, genre, release decade, number of valid titles, average rating, minimum rating, and maximum rating — all separated by commas. Here is a screenshot of the output file:



```
part-r-00000
File Edit View H1  B I S G  A
Movie,Action & Adventure,1910s ,1,5.71,5.7,5.7
Movie,Action & Adventure,1920s ,1,6.91,6.9,6.9
Movie,Action & Adventure,1930s ,4,5.96,4.9,6.5
Movie,Action & Adventure,1940s ,5,6.11,4.5,6.9
Movie,Action & Adventure,1950s ,2,5.76,4.9,6.6
Movie,Action & Adventure,1960s ,1,4.21,4.2,4.2
Movie,Action & Adventure,1970s ,3,6.64,5.9,7.4
Movie,Action & Adventure,1980s ,10,5.67,4.1,7.6
Movie,Action & Adventure,1990s ,15,5.68,3.7,7.1
Movie,Action & Adventure,2000s ,16,5.70,3.2,7.8
Movie,Action & Adventure,2010s ,39,5.45,1.7,8.2
Movie,Action & Adventure,2020s ,7,3.44,1.4,5.7
Movie,Animation,1940s ,1,6.31,6.3,6.3
Movie,Animation,1950s ,2,7.46,7.4,7.5
Movie,Animation,1970s ,2,6.91,6.7,7.1
Movie,Animation,1980s ,1,8.61,8.6,8.6
Movie,Animation,1990s ,4,5.84,3.6,7.1
Movie,Animation,2000s ,7,7.15,6.2,8.6
Movie,Animation,2010s ,19,7.04,5.5,8.5
Movie,Animation,2020s ,5,6.81,5.6,7.9
Movie,Biography,1950s ,1,5.91,5.9,5.9
Movie,Biography,1980s ,1,8.21,8.2,8.2
Movie,Biography,1990s ,4,6.96,5.5,7.8
Ln 1, Col 1 6,525 characters Plain text
```